

Methods: BHR_71, HH_212, IRAS_0416+2706, L1551_IRS_5,
CG_30, HH_46, GAL_331.5-00.1, IRAS_04166+2706

ALMA Thesis Planner (automated pipeline)

0.1 0. Framing and ground rules

You are going to turn six archival ACA tunings on a single field into (i) a verified line inventory and (ii) a three-zone chemical contrast measurement. Everything below is organized so that the field-of-view and noise properties are established *before* any science aperture is chosen, because on a single 7-m pointing the honest usable area is much smaller than the outflow, and every subsequent number depends on knowing where the data are trustworthy. Keep a bound lab notebook — a dated Markdown log in the repository is fine — and record every CASA task call, parameter, and output filename as you go. You will not remember in April what ‘imsmooth’ kernel you used in October.

0.2 1. Data retrieval, inventory, and provenance

Download from the ALMA Science Archive (<https://almascience.nrao.edu/aq/>), searching project code ‘2017.1.01053.S’, and select only the six BHR 71 member OUSs:

MOUS	Nominal windows (GHz)	Δv (km/s)	Beam (")	— — —	‘uid://A001/X1288/X1001’
292.7–294.7, 294.6–296.6, 304.7–306.7, 306.5–308.5	1.104	4.147		‘uid://A001/X1288/Xfee’	
296.4–298.4, 298.3–300.3, 308.4–310.4, 310.3–312.3	1.091	4.087		‘uid://A001/X1288/Xfdb’	
300.2–302.2, 302.0–304.0, 312.1–314.1, 314.0–316.0	1.078	3.997		‘uid://A001/X1288/Xfc8’	
337.7–339.7, 339.6–341.6, 349.7–351.7, 351.5–353.5	0.963	3.606		‘uid://A001/X1288/Xfb5’	
341.5–343.4, 343.3–345.3, 353.4–355.4, 355.3–357.3	0.953	3.545		‘uid://A001/X1288/Xfa2’	
345.2–347.2, 347.1–349.1, 357.1–359.1, 359.0–361.0	0.943	3.526			

Pull both the pipeline-calibrated measurement sets and the QA2 image products (‘*pbcor.fits*’, the *un-pbcor* ‘*image*’, the ‘*pb*’, and the ‘*psf*’/beam tables). Read the QA2 report and the ‘weblog’ for each MOUS before touching a pixel: note the flux calibrator used, the reported flux-scale uncertainty, any flagged antennas, and the achieved vs. requested sensitivity. Tabulate for each of the 24 spectral windows: sky frequency range, channel width in MHz and* km/s, number of channels, restoring beam major/minor/PA, and the pipeline’s reported RMS. This table is Table 1 of your thesis and the reference for every later step.

Verify the pointing centres agree across the six MOUSs to within a fraction of a beam (they are quoted as 180.40337/8, -65.14701) and confirm ‘Mosaic = no’ in the headers — you are working with a single pointing, and the entire analysis is built around that fact.

Decision on re-imaging. Default to the QA2 cubes. Re-run ‘tclean’ yourself only for a specific, stated reason: (a) if the QA2 continuum subtraction (‘uvcontsub’) removed real line flux in a line-crowded window, or (b) if you need a channelization or a restoring beam the pipeline did not deliver.

If you do re-image, use ‘specmode=’cube‘, ‘outframe=’LSRK‘, ‘gridder=’standard‘ (single field), ‘deconvolver=’hogbom‘, ‘weighting=’briggs‘ with ‘robust=0.5‘, and clean to a threshold of $\sim 2\times$ the measured per-channel RMS inside a hand-drawn mask. Document that you did it and show a QA2-vs-yours spectrum comparison at the continuum peak so the reader knows nothing moved.

0.3 2. Step one: the primary-beam-corrected noise map and the usable radius

Do this before you choose apertures, before you make moment maps, before anything.

For each MOUS:

1. From the ‘pb’ image, confirm the primary-beam response. For a 7-m dish, $\text{FWHM} \approx 1.13 \lambda/D$, i.e. $\approx 33\text{--}34''$ at 300 GHz and $\approx 29''$ at 345 GHz. Verify this against the actual ‘pb’ cube rather than assuming it, and note that the PB is frequency-dependent *within* each 2 GHz window (a $\sim 0.6\%$ effect, negligible) and *between* the 300 and 350 GHz groups (a $\sim 15\%$ difference in FWHM, not negligible).
2. Measure σ_0 , the RMS in the flat-noise (un-pbcor) cube, per channel, in signal-free channels and signal-free sky regions. Use a sigma-clipped standard deviation (‘astropy.stats.sigma_clipped_stats‘, 3σ , 5 iterations) over a spatial annulus you have visually confirmed to be emission-free, and repeat channel-by-channel so you have $\sigma_0(\nu)$. Plot σ_0 against channel for each spw; this immediately shows you where the bandpass edges degrade and where bright lines are contaminating your “signal-free” selection.
3. Build the noise map: $\sigma(x, y, \nu) = \sigma_0(\nu) / \text{PB}(x, y, \nu)$. Write this out as a FITS cube alongside each science cube. Every uncertainty you quote for the rest of the thesis comes from this cube — never from the tabulated 8.4–13.6 mJy/beam values in the proposal metadata, which apply only at the pointing centre.
4. Convert to a per-10-km/s and a per-channel sensitivity so you can compare to the archive numbers as a sanity check at $r = 0$.
5. Define and state the **usable radius** r_{max} : the radius at which $\text{PB} \geq 0.5$ (i.e. noise $\leq 2\times$ the central value) for the *highest* frequency tuning, since that is the most restrictive. Expect $r_{\text{max}} \approx 14\text{--}15''$. Draw this circle on every finding chart in the thesis. **No science aperture may be centred outside it, and no upper limit is quoted from outside it.** If you are ever tempted, remember that the flux calibration and the PB model both degrade fastest exactly where the emission is faintest.

Deliverable: one figure showing PB-corrected RMS vs. radius for all six tunings, one stated r_{max} , and a one-paragraph justification.

0.4 3. Continuum images

Only cubes are archived, so build the continuum yourself, per tuning.

Work on the flat-noise cubes. For each spw, identify line-free channels by an iterative sigma-clip on the spatially-integrated spectrum inside r_{max} : compute the mean spectrum, clip channels deviating $>2.5\sigma$ from a running median, refit, iterate to convergence. Toward a chemically rich source in an unbiased scan, expect that a large fraction of channels are contaminated; that is a result, not a failure. Then either

- fit and subtract a low-order (0th or 1st) polynomial baseline per pixel over the surviving channels using ‘imcontsub‘, taking the fitted continuum image as your product; or
- run ‘uvcontsub‘ on the MS with the same channel selection and ‘tclean‘ the continuum with ‘specmode=’mfs‘.

Do it both ways for at least one tuning and confirm agreement to within the noise.

Report the fraction of channels used per spw. Then quantify the residual line contamination: take the *rejected* channels, compute their mean excess brightness over the fitted continuum at the peak, and multiply by the fraction of the band they occupy. That number is your systematic on the continuum flux, and you quote it explicitly, because the continuum peak defines aperture (a). Compare the six per-tuning continuum peak positions; they should agree to well within a beam. Their scatter is your empirical astrometric uncertainty on the aperture centre.

Fit a 2D Gaussian ('imfit') to the continuum peak to get position, deconvolved size, peak, and integrated flux, with the PB-corrected local RMS as the input error.

0.5 4. Homogenising the data to a common beam

The BHR 71 beams span 3.526''–4.147''. Set the common circular restoring beam at **4.2''** — driven by this field, not by the other SMORES targets.

For every cube, in this order:

1. 'imsmooth(kernel='gauss', targetres=True, major='4.2arcsec', minor='4.2arcsec', pa='0deg')' on the **flat-noise** cube. Smoothing before PB correction keeps the noise statistics tractable.
2. Primary-beam correct by dividing by the 'pb'.
3. Convert Jy/beam $\rightarrow$ K using the Rayleigh–Jeans relation, $T_b = 1.222 \times 10^3 \times I[\text{Jy/beam}] / (\nu[\text{GHz}]^2 \theta_{\text{maj}}[\text{''}] \theta_{\text{min}}[\text{''}])$, with $\theta = 4.2''$. Use 'spectral-cube's 'to(u.K)' and cross-check one channel by hand against the formula.
4. Regrid all cubes onto a single common spatial grid with 'imregrid' so that pixel (i, j) is the same sky position in all 24 spws.
5. Propagate the noise map through the same operations. Smoothing correlates noise: re-measure σ_0 empirically on the *smoothed* cube rather than scaling analytically, then re-divide by the PB.

Sanity check: after homogenisation, extract the CO 3–2 spectrum at the continuum peak from the smoothed and unsmoothed cubes; the integrated intensity should be preserved to within a few percent.

Also convert all spectral axes to LSRK velocity ('imreframe', 'outframe='LSRK') and adopt a single systemic velocity (Section 7) for all subsequent velocity work.

0.6 5. The line inventory — a standalone deliverable

Treat this as a self-contained chapter that stands whether or not any fit later succeeds.

1. Extract a reference spectrum in a fixed aperture (one 4.2'' beam at the continuum peak) from every one of the 24 spws, and stitch them into a continuous 292.7–316.0 + 337.7–361.0 GHz scan in T_b. Plot it as a multi-panel figure at a fixed vertical scale, then again zoomed.
2. Identify every feature above 4σ (using the local noise from Section 2) with an automated peak-finder, then verify **each one by hand in Splatalogue** (or via 'astroquery.splatalogue'), querying CDMS and JPL. **The line list in the data description is LLM-generated and is a starting hypothesis only** — you re-derive every rest frequency and E_up from the catalogues yourself and cite the catalogue entry.
3. Apply consistent identification criteria and state them: the line must fall within ± 2 channels of its catalogued rest frequency after applying v_sys; the species must have at least one other transition in the covered bands consistent in intensity with the same excitation conditions (or, if it is the only accessible transition, say so); and blends must be flagged where two catalogued transitions fall within one resolution element.
4. Produce the master table: species, quantum numbers, rest frequency, E_up, $\log_{10}(A_{ij})$ or $S \mu^2$, catalogue, which tuning/spw it lands